

An explicit 64-modular Hadamard matrix of order 892

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Abstract

We construct a sign matrix H of order 892 satisfying $HH^T \equiv 892I_{892} \pmod{64}$. Three subsets of $\mathbb{F}_{223}$, each consisting of zero and one multiplicative coset of size 37, have total difference multiplicity 19 at every nonzero shift. Together with a constant sequence, they give a length-223 quadruple whose total periodic autocorrelation is 512 away from zero. An explicit Goethals–Seidel array then satisfies the exact identity $HH^T = 380I_{892} + 512(I_4 \otimes J_{223})$. We give the complete finite intersection certificate, the block identity, and an exactly checkable serialized matrix. No claim of historical priority or of an ordinary Hadamard matrix is made.

1 Statement and scope

A sign matrix has all its entries in $\{-1, 1\}$. An m -modular Hadamard matrix of order n is an $n \times n$ sign matrix H for which $HH^T \equiv nI_n \pmod{m}$, entrywise. Write I_n for the identity matrix and J_n for the all-one matrix of order n ; $\otimes$ denotes the Kronecker product. The question addressed here is precisely whether such a matrix exists for $(n, m) = (892, 64)$.

Theorem 1. *The matrix H defined in (6) is a sign matrix of order 892 and satisfies*

$$HH^T = 380I_{892} + 512(I_4 \otimes J_{223}). \quad (1)$$

In particular, $HH^T \equiv 892I_{892} \pmod{64}$.

The contribution is the fully specified matrix and a finite proof of (1). The proof combines multiplicative cosets with a circulant block array; all required identities are established below. As broad methodological context, coset constructions in finite groups appear in Zeng’s work on rainbow trees [1], while exact matrix certificates appear in his work on rational Hadamard factorization [2]. Those papers concern different problems; the latter uses the entrywise Hadamard product, rather than modular orthogonality. Neither is used as a premise here. The relationship of the present supports to earlier cyclotomic and modular Hadamard constructions has not been established. We therefore make no novelty, priority, or smallest-previously-unresolved-order assertion.

2 Three supports and their differences

All residues and shifts in this section are taken modulo 223. Trial division by 2, 3, 5, 7, 11, 13 verifies that 223 is prime. Define

$$S = \{2^r : 0 \leq r < 37\} \subset \mathbb{F}_{223}^\times, \quad C_j = 3^j S, \quad X_j = \{0\} \cup C_j \quad (j = 3, 4, 5). \quad (2)$$

The congruence $2^{37} = 1$ and the primality of 37 show that S is a multiplicative subgroup of order 37. Moreover,

$$3^{111} = -1, \quad 3^{74} = 183, \quad 3^6 = 60.$$

These three residues differ from 1; since the prime divisors of 222 are 2, 3, 37, the order of 3 is 222. Thus $C_0, \dots, C_5$ partition $\mathbb{F}_{223}^\times$, and $-C_j = C_{j+3}$, with subscripts modulo 6. Here $\langle 3^6 \rangle = S$, as both are the subgroup of order 37 in the cyclic group generated by 3.

For an explicit certificate, the three supports are

$$\begin{aligned} X_3 &= \{0, 13, 26, 27, 52, 54, 59, 87, 91, 95, 103, 104, 108, 111, 118, 125, \\ &\quad 141, 155, 157, 159, 163, 167, 174, 182, 189, 190, 191, 193, 195, \\ &\quad 206, 207, 208, 209, 215, 216, 219, 221, 222\}, \\ X_4 &= \{0, 19, 25, 31, 38, 39, 43, 50, 55, 62, 76, 78, 81, 86, 89, 100, 101, \\ &\quad 110, 121, 124, 127, 131, 133, 139, 152, 156, 162, 172, 175, 177, \\ &\quad 178, 181, 199, 200, 202, 211, 217, 220\}, \\ X_5 &= \{0, 5, 10, 11, 20, 22, 35, 40, 44, 57, 70, 75, 77, 79, 80, 85, 88, 93, \\ &\quad 97, 107, 114, 117, 129, 140, 149, 150, 151, 154, 158, 160, 165, \\ &\quad 170, 176, 186, 187, 194, 205, 214\}. \end{aligned}$$

Each has 38 elements. For $j \in \{3, 4, 5\}$, put

$$N_j(t) = |\{x \in X_j : x + t \in X_j\}|.$$

Lemma 1. *For every $t \in \mathbb{F}_{223} \setminus \{0\}$,*

$$N_3(t) + N_4(t) + N_5(t) = 19. \quad (3)$$

Proof. Multiplication by $s \in S$ maps each X_j bijectively to itself, whence $N_j(st) = N_j(t)$. Reversing an ordered pair gives $N_j(-t) = N_j(t)$; this does not require $X_j = -X_j$. Consequently it suffices to check $t = 1, 3, 9$, representing the three pairs of cosets (C_0, C_3) , (C_1, C_4) , and (C_2, C_5) . Each pair contains 74 nonzero residues, and together they cover all 222 nonzero shifts. Table 1 lists every contributing x at these representatives. Each row is checked by testing the 38 elements of the displayed support. The three totals are $10 + 4 + 5$, $5 + 10 + 4$, and $4 + 5 + 10$, respectively, proving (3). $\square$

t	j	All $x \in X_j$ with $x + t \in X_j$	$N_j(t)$
1	3	26, 103, 189, 190, 206, 207, 208, 215, 221, 222	10
1	4	38, 100, 177, 199	4
1	5	10, 79, 149, 150, 186	5
3	3	108, 190, 206, 216, 219	5
3	4	78, 86, 121, 124, 172, 175, 178, 199, 217, 220	10
3	5	77, 85, 114, 151	4
9	3	95, 182, 206, 207	4
9	4	101, 124, 172, 202, 211	5
9	5	11, 35, 70, 79, 88, 140, 149, 151, 205, 214	10

Table 1: Complete intersection certificates; addition is modulo 223.

3 The sequences and the block matrix

For $j = 3, 4, 5$ let $q_j(i) = 2\mathbf{1}_{X_j}(i) - 1$. Use the ordered quadruple

$$(a, b, c, d) = (\mathbf{1}, q_5, q_4, q_3), \quad (4)$$

where $\mathbf{1}(i) = 1$ for every $i \in \mathbb{F}_{223}$. For any such sequence u , define its periodic autocorrelation by

$$P_u(t) = \sum_{i \in \mathbb{F}_{223}} u(i)u(i+t).$$

Expanding the two indicator functions, and using $|X_j| = 38$, gives

$$P_{q_j}(t) = 223 - 4 \cdot 38 + 4N_j(t) = 71 + 4N_j(t).$$

Since $P_a(t) = 223$, Lemma 1 yields

$$P_a(t) + P_b(t) + P_c(t) + P_d(t) = \begin{cases} 892, & t = 0, \\ 512, & t \neq 0. \end{cases} \quad (5)$$

The value at zero follows separately because each sequence has 223 sign entries.

Index rows and columns of the following blocks by $0, \dots, 222$. Let $A_{ij} = a(j-i)$, and define B, C, D similarly from b, c, d . Let $R_{ij} = 1$ if $i+j = 222$, and $R_{ij} = 0$ otherwise. Set

$$H = \begin{pmatrix} A & BR & CR & DR \\ -BR & A & -D^\top R & C^\top R \\ -CR & D^\top R & A & -B^\top R \\ -DR & -C^\top R & B^\top R & A \end{pmatrix}. \quad (6)$$

This formula fixes all 892^2 entries, including the order of the sequences and the reversal convention.

Proof of Theorem 1. The blocks of H are signed column permutations of sign circulants or their transposes, so H is a sign matrix. Circulants and their transposes commute: each is a polynomial in the cyclic shift matrix. Furthermore, $R^\top = R$, $R^2 = I$, and

$$RX = X^\top R \quad (7)$$

for any circulant $X_{ij} = x(j-i)$. Indeed, both sides of (7) have (i, j) entry $x(i+j+1)$, with the argument reduced modulo 223.

Each diagonal block of $HH^\top$ is

$$Q = AA^\top + BB^\top + CC^\top + DD^\top.$$

This uses $XX^\top = X^\top X$ for circulant X . For completeness, the six upper off-diagonal blocks reduce to

$$\begin{aligned} (HH^\top)_{12} &= (-AB + BA)R - CD + DC, \\ (HH^\top)_{13} &= (-AC + CA)R + BD - DB, \\ (HH^\top)_{14} &= (-AD + DA)R - BC + CB, \\ (HH^\top)_{23} &= BC^\top - C^\top B + (AD^\top - D^\top A)R, \\ (HH^\top)_{24} &= BD^\top - D^\top B + (-AC^\top + C^\top A)R, \\ (HH^\top)_{34} &= CD^\top - D^\top C + (AB^\top - B^\top A)R. \end{aligned}$$

Here the subscripts denote block indices. All six expressions vanish by commutativity; the lower blocks are their transposes. Hence $HH^\top = I_4 \otimes Q$.

An entry $(AA^\top)_{ij}$ is $P_a(i - j)$, and similarly for the other circulants. Equation (5) therefore gives $Q = 380I_{223} + 512J_{223}$, proving (1). Finally,

$$HH^\top - 892I_{892} = 512[I_4 \otimes (J_{223} - I_{223})],$$

which is entrywise divisible by 64. □

4 Exact verification and limitations

The accompanying `matrix-892.txt` stores 892 rows of 892 ASCII signs + and -, each followed by a line feed. The supplementary standard-library Python program `verify.py` reconstructs the supports and all matrix entries from (2)–(6), compares them with this file, and checks (1) on the serialized signs. From the supplementary directory it is run as

```
python3 verify.py matrix-892.txt.
```

If two sign rows differ in d positions, their inner product is $892 - 2d$. This gives an exact integer test using binary encodings of the rows. All $397\,386 = \binom{892}{2}$ unordered distinct row pairs and all 892 diagonal entries are checked; symmetry covers all $795\,664$ ordered Gram entries. Among distinct unordered pairs, $99\,012$ have inner product 512, and $298\,374$ have inner product zero.

The construction was located by a finite search over sequences constant on multiplicative cosets. The proof uses only the displayed witness and finite arithmetic, and requires no assertion about search completeness. Separate exact verification implementations checked the support differences, all 222 nonzero autocorrelation sums, and the full Gram identity. A separate model-context check reconstructed the matrix without importing the search implementation; a single-sign corruption was also rejected. These checks use exact integers, not a floating-point tolerance. They supplement the finite proof and do not constitute proof-assistant formalization or external peer review.

Equation (1) has nonzero off-diagonal entries, so the exhibited matrix is not an ordinary real Hadamard matrix. No general family theorem, inequivalence classification, or historical status claim follows from this one construction. In particular, the present note does not determine whether the same supports or an equivalent matrix occurred in earlier work.

Computational and AI assistance. OpenAI Codex assisted with the finite search, the proof exposition, verification programs, a separate model-context mathematical review, and manuscript preparation. The supplied matrix and deterministic checks make the finite calculation reproducible; model-generated assessments are not used as mathematical premises.

References

- [1] Zijian Zeng. *Coset and Fiber Constructions for Rainbow Trees in Finite Abelian Groups*. 2026. <https://doi.org/10.2139/ssrn.7387259>.
- [2] Zijian Zeng. *A Rational Rank-(2, 2) Hadamard Factorization of a Proposed 4×4 Real Counterexample*. Preprint, Zenodo, September 2026. <https://doi.org/10.5281/zenodo.22242517>.